

Parametric Evaluation of KNSB Solid Rocket Propellant

Subject: Part 2 — Modification of KNSB via Suppressants to Lower the Thrust Curve

Source Data: Richard Nakka's Experimental Rocketry Database

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2. Modification of KNSB via Suppressants to Lower the Thrust Curve

In solid rocket motor design, tailoring the thrust-time profile is necessary to meet key structural and flight envelopes. While burn rate catalysts focus on maximizing prompt booster acceleration, burn rate suppressants are intentionally integrated into the Potassium Nitrate/Sorbitol (**KNSB**) matrix to flatten the thrust curve, minimize Maximum Expected Operating Pressure (MEOP), and extend core burning time for sustainer phases.

2.1 Chemical Suppressants: Sodium Bicarbonate and Magnesium Sulfate

Chemical suppressants decrease core regression velocity by introducing clean, endothermic decomposition loops that absorb thermal energy directly out of the propellant's primary reaction boundary.

- **Sodium Bicarbonate (NaHCO_3 / Baking Soda):** Nakka's empirical testing indicates that Sodium Bicarbonate serves as a highly robust burn rate depressant. When exposed to the primary combustion wave, it undergoes rapid endothermic decomposition, liberating inert carbon dioxide (CO_2) gas and water vapor (H_2O). This gas release effectively dilutes volatile gas-phase intermediates and cools the local flame zone, lowering the linear burn rate. Nakka utilizes a 5.0% mass loading of sodium bicarbonate within specialized tracking smoke grains to suppress energetic flame fronts, forcing the composition to cleanly smolder and vaporize secondary smoke agents (such as paraffin wax) without accelerating into open combustion.
- **Magnesium Sulfate (MgSO_4 / Epsom Salts):** Doping the formulation with minor fractions (such as 1.0% mass inclusion of anhydrous MgSO_4) acts as a physical burn rate depressor. Its presence interrupts clean heat transfer to the unburned solid face, depressing the localized regression rate noticeably within lower chamber pressure environments.

2.2 Stoichiometric Modifications: Fuel-Rich O/F Shifting

The standard, optimized baseline formulation for KNSB relies on a 65% Potassium Nitrate and 35% Sorbitol ratio (an Oxidizer-to-Fuel ratio of 1.86). Shifting this chemical balance toward a

fuel-rich profile (reducing the split to 60/40 or 58/42) acts as a structural suppressant mechanism:

- **Combustion Temperature Drop:** Introducing excess fuel leads to incomplete oxidizer bonding, shifting the chemical equilibrium away from carbon dioxide production toward carbon monoxide (**CO**), which significantly decreases core flame temperatures.
- **Kinetic Rate Suppression:** Lowered flame temperatures directly limit the radiant and conductive heat flux returning to the solid propellant grain face, slowing the solid-to-gas phase change and extending total thrust duration.
- **Rheological Advantage:** This stoichiometric shift lowers the dynamic viscosity of the melted propellant slurry, transforming a dense mix into a highly fluid melt that helps eliminate structural air void defects during casting.

2.3 Physical Suppressants: Particle Size Dynamics

Solid-state propellants burn via heterogeneous surface reactions. Modifying the initial particle sizing of the **KNO₃** crystal alters the combustion velocity without forcing changes to the foundational chemistry:

- **Fine-Grind Oxidizer (25–200 microns):** Milled extensively in an electric grinder to a flour-like consistency. This maximizes contact surface boundaries with the sorbitol binder, yielding high combustion efficiency, high internal pressure spikes, and prompt thrust response.
- **Coarse-Grind Oxidizer (100–800 microns):** Unrefined, granular crystals resembling table salt. Utilizing un-milled crystals narrows the initial surface contact boundaries. This acts as a physical burn rate suppressant, lowering the burn rate coefficient, flattening peak chamber pressures, and stretching out the thrust profile over a longer time horizon.

2.4 Ballistic Exponent Variations and Mesa Burn Behavior

The burning rate behavior of KNSB is mathematically defined via Saint Robert's power law:

$$r = a * P^n$$

Where **r** represents the linear regression burn rate, **a** is the empirical burn rate coefficient, **P** is the local combustion chamber pressure, and **n** is the pressure exponent. The pressure exponent **n** dictates how sensitive the burn velocity is to localized pressure changes; high values introduce risks of rapid over-pressurization loops.

Strand burner and ballistic evaluation motor testing by Richard Nakka confirms that KNSB deviates from standard linear power models, displaying distinct **Mesa burn behavior**. Within a distinct mid-pressure envelope, the burn rate plateaus or decreases slightly as pressure climbs, providing an inherent safety margin against progressive over-pressurization. The addition of

physical and chemical suppressants selectively shifts these parameters across operating regimes.

3. Analytical Reference Data & Ballistic Matrices

The tables below outline the empirical burn rate coefficients and pressure exponents for standard fine-grind KNSB across specific pressure boundaries, followed by a comparative matrix of suppression mechanisms.

Table 3.1: KNSB Burn Rate Parameters Across Pressure Boundaries

Propellant State / Regime	Pressure Range (MPa)	Pressure Range (psi)	Exponent (n)	Coefficient (a)*
Standard KNSB (Low Pressure)	0.10 to 1.38	14.7 to 200	0.684	5.131
Standard KNSB (Mid Pressure / Mesa Regime)	1.38 to 3.79	200 to 550	0.311	8.835
Standard KNSB (High Pressure Range)	3.79 to 7.03	550 to 1020	0.535	3.907
KNSB (Coarse-Grind Suppressed)	0.10 to 7.03	14.7 to 1020	Variable	~94% of Fine Baseline

**Note: Units for burn rate coefficient 'a' are formulated to yield burn rates in mm/sec when chamber pressure is calculated in MPa.*

Table 3.2: Comprehensive Matrix of Suppressant Modifications

Suppressant Type	Typical Loading	Primary Ballistic Mechanism	Impact on Thrust Curve	Key Processing Trade-Offs
Sodium Bicarbonate (NaHCO ₃)	1.0% to 5.0% mass	Endothermic chemical decomposition; liberates CO ₂ and H ₂ O to cool the flame front.	Substantially lowers the curve; dampens peak pressure spikes.	Can make ignition difficult; requires a highly energetic igniter/starter layer.
Fuel-Rich Shifting (O/F reduction)	60/40 to 58/42 split	Alters stoichiometric balancing, shifting exhaust species to CO and lowering thermal energy.	Flattens the profile; extends the total burn duration.	Slightly lowers total Specific Impulse; improves melted slurry fluidity.
Coarse-Grind Oxidizer	100% of KNO ₃ mass	Physical restriction of reaction boundaries due to reduced particle surface area.	Reduces initial thrust spike; shifts peak pressure outward.	Introduces slight combustion inefficiency; casting slurry requires manual packing.
Magnesium Sulfate (MgSO ₄)	1.0% doping	Physical thermal absorption and obstruction of solid-state heat flux.	Depresses regression rate, primarily in low-pressure regimes.	Must be entirely anhydrous to prevent moisture-induced mechanical grain soft spots.

VERIFIED REFERENCE LITERATURE

This technical report synthesizes empirical findings, raw data coordinates, and safety methodologies derived directly from the laboratory records of Richard Nakka's Experimental Rocketry database:

- **Oxidizer Modifiers & Depressants:** nakka-rocketry.net/oxidex.html
- **Burn Rate Instrumentation & Strand Testing:** nakka-rocketry.net/bntest.html
- **Propellant Ballistics & Regression Physics:** nakka-rocketry.net/burnrate.html